

Find a Real Problem

Project: Helpful Classroom Sorter

1. Find a Real Problem

Teach the learning cycle

Primary target: Describe a specific user difficulty using supplied evidence without inventing its cause

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system's capability, the correctness of its output and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time. The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Prompts Are Design Instructions

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt with a clear task, supplied evidence and checkable output requirements

Reference example, not the only acceptable wording: Possible omissions include audience, source facts, required fields, format, missing-value behaviour and limits on extra claims. C supplies evidence, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: $9/12 = 3/4 = 75\%$. The denominator includes all 12 judged outcomes, including 3 incorrect ones. Using 9/9 would exclude the errors and falsely report perfect assessed accuracy.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Students spend three minutes finding returned books. Compare shelf labels with an AI locator. Define one fair success test before choosing a solution.

Concept to explain

A strong project begins with a specific user and observable need.

Term · Meaning

Term: user

Meaning: person affected by the problem or solution

Term: need

Meaning: observable difficulty or goal

Term: interview

Meaning: structured questions used to learn from a person

Term: success measure

Meaning: evidence showing whether the need improved

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This begins an eight-week project sequence culminating in an evidence-based defence. Use the activity recommender as a shared paper project; later weeks specify its catalogue and tests. The supplied counts are fictional motivation, not collected class data or a result of an AI system. Assess problem framing separately from enthusiasm for a particular technology. A justified decision to retain a simple method is a valid project conclusion.

Readiness answer

No. The permission covers a paper prototype and fictional requests. Collecting and publishing personal choices are different actions outside that scope.

The fictional ten-request activity observation in the lesson

Paper for a brief with user, evidence, outcome, inputs and boundaries

The independent equipment-cupboard scenario in W4

End goal for the pictorial case

By the end, I can turn a recorded problem into a bounded, measurable project goal.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT observations and interview evidence

PROCESS identify user, situation, need, and measurable effect

OUTPUT testable problem statement without a preselected solution

CORE IDEA A strong project begins with a specific user and observable need.

UNDER THE HOOD Problem framing separates evidence about a user's need from assumptions about a solution. A strong statement identifies who, when, what observable difficulty occurs, and how improvement could be measured.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE 'Students need AI' is not a problem statement; 'students spend five minutes finding misplaced supplies after art' is observable and testable.

STEP 1 — OBSERVE observations and interview evidence

STEP 2 — TRACE identify user, situation, need, and measurable effect

STEP 3 — CONCLUDE testable problem statement without a preselected solution

MODEL REASONING A strong statement names the user, context, observable difficulty, and desired improvement without requiring AI. Protect privacy, avoid leading questions, and distinguish what the user said from your inference.

DO NOT OVERCLAIM Do not collect personal stories unrelated to the design question.

What new evidence could change this answer?

Concrete teaching case

Shared markers are often lost because tray labels are unclear.

State the need before selecting AI.

Exact explanation to model

Students need to return markers correctly and quickly. Test clearer labels before proposing a model.

Explain the added worked example

For the final project sequence, we will develop a small activity recommender on paper. Its users are fictional pupils choosing a classroom activity that fits their available time and room conditions. We will decide what the recommender must do, build a traceable process, test it and explain its limits. A recommendation is a suggestion; the pupil can decline it.

Start before the solution. Our supplied observation log describes ten fictional requests made using an activity catalogue. Six first choices fit the request's time and room conditions. Four did not and needed another choice. These invented classroom records give us a problem to investigate. They are not observations of your classmates and do not show how often this happens in other classes.

A useful problem statement is: "In the supplied ten-request exercise, four first choices did not fit the stated time or room conditions, so those users had to choose again." "We need AI" names a possible technology but says neither who needs help nor what should improve. The log also does not establish whether unclear wording, missing information or another cause produced the mismatches.

Our target outcome is a compatible suggestion when the catalogue contains one, and a clear no-match message when none fits. Compatibility means respecting the stated conditions; it does not mean that every pupil will enjoy the activity. We can check time and room conditions from a request and catalogue. Enjoyment would need different evidence.

Count compatible suggestions, incompatible suggestions and no-match messages separately. Otherwise a system could avoid every wrong suggestion by refusing to suggest anything. If a valid option existed, that refusal would still fail to provide the intended help. Record the total number of requests so a count has a clear denominator.

Key terms

Term · Meaning

Term: User need

Meaning: Something a person needs to accomplish in a particular situation

Term: Problem statement

Meaning: A specific description of who faces a difficulty and what that difficulty is

Term: Outcome

Meaning: The result experienced by the user

Term: Constraint

Meaning: A condition that a proposed solution must respect

Term: Success measure

Meaning: An observable quantity or result used to judge progress

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Users and setting · Fictional classroom activity requests · A clear situation to test

Users and setting: Information

Fictional classroom activity requests: Available minutes and quiet-room requirement

A clear situation to test: Relevant to compatibility

Users and setting: Action

Fictional classroom activity requests: Suggest an activity or report no match

A clear situation to test: The user retains the choice

Users and setting: Outside the project

Fictional classroom activity requests: Names, profiles, purchases and public posting

A clear situation to test: Not needed for this paper task

The teacher permits a paper prototype using fictional requests and an approved catalogue. We do not need names, home details or a history of a pupil's choices. A request can say "10 minutes, quiet room required" without identifying anyone. Use only the request information needed to test the intended outcome.

A clear catalogue or a short checklist may solve this problem. We will compare a proposed recommender with a simple starting method instead of assuming a more complicated system is better. This project can use explicit rules; it does not have to fit a machine-learning model. The choice should follow the problem, available evidence and limits of the task.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Find a Real Problem

A fictional class loses time finding the correct craft tray.

The facts we will use

In eight recorded requests, three selected trays lack a required tool; five match.

The teacher permits a paper finder using fictional requests and an approved tray list, with no pupil names.

Find a Real Problem

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe problem

Three mismatches are recorded out of eight requests.



2. Define need

Help users choose a tray containing every requested tool.



3. Set success test

Compare selected tray against the approved inventory for each new case.



4. Bound the project

Use fictional tool requests; do not collect identity or purchase equipment.

CHANGE ONE THING

The approved list contains no tray with every requested tool.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Observe problem

What happens and why: Three mismatches are recorded out of eight requests.

Step: 2. Define need

What happens and why: Help users choose a tray containing every requested tool.

Step: 3. Set success test

What happens and why: Compare selected tray against the approved inventory for each new case.

Step: 4. Bound the project

What happens and why: Use fictional tool requests; do not collect identity or purchase equipment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The mismatch record describes a concrete need. A measurable goal checks whether the chosen tray satisfies the request; calling the system smart would not tell us how to evaluate it. Start with a transparent paper baseline, and keep unsatisfied requests visible instead of selecting a near match without warning.

What this example does not establish

Eight requests do not show the problem's frequency across the entire school or prove AI is necessary.

Change one thing in the pictured case

Change: The approved list contains no tray with every requested tool.

Prediction: The finder should report no suitable tray rather than guess.

Reason: Success includes honestly identifying unsatisfied requirements.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Shared markers are often lost because tray labels are unclear.

Write the question to explain: State the need before selecting AI.

Reveal after predictions: Students need to return markers correctly and quickly. Test clearer labels before proposing a model.

Keep for an independent attempt: Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The brief should identify fictional pupils choosing classroom activities, four incompatible first choices out of ten, and the goal of a compatible suggestion where one exists or an explained no-match result otherwise. Inputs are minutes and quiet-room requirement. Boundaries include fictional requests, teacher-approved catalogue and suggestions only. A clear catalogue or checklist is a simpler alternative. The log does not establish causes, feelings or a future tool's benefit. Exit: no; always asking avoids a wrong suggestion but may miss available compatible activities. Check useful matches and justified no-match cases, not only avoided errors.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve the limits of permission for a paper prototype.

Stage: 7 to 20 minutes

Teacher action: Read the observation and distinguish difficulty, cause and proposed solution. Complete W1.

Stage: 20 to 33 minutes

Teacher action: Trace the evidence-to-outcome diagram and complete W2.

Stage: 33 to 45 minutes

Teacher action: Select relevant inputs and a simple alternative in W3.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve why four mismatches do not prove a cause or a solution.

Stage: 8 to 23 minutes

Teacher action: Draft the guided project brief; challenge unsupported claims and excess data.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently, checking the brief against the new scenario.

Stage: 38 to 45 minutes

Teacher action: Complete W5 and the exit; discuss how a comparison could favour the simple method.

Reduce help deliberately

Completion task: A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

Teacher check: The log establishes four incompatible first choices out of ten supplied fictional requests. It does not establish every pupil's opinion, the cause of the mismatches or that an untested recommender will improve outcomes. Accept any two of those unsupported conclusions with the bounded observation.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for A problem worth solving. Proposed explanation: "Using AI is itself the outcome". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: State what improves for the user and how to observe it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Write a one-sentence project goal and say what evidence would count as a failure.

Reasoned teacher answer

Goal: choose an inventory-confirmed tray containing all requested tools or report none. A selected tray missing a required tool is a failure.

Changed-case reasoning: The finder should report no suitable tray rather than guess. Success includes honestly identifying unsatisfied requirements.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Interview a classmate and write one testable need statement.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Every classroom problem needs AI.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Using AI is itself the outcome: State what improves for the user and how to observe it.

A mismatch count proves its cause: Separate observations from explanations that need further evidence.

No wrong suggestions means success: Check whether valid options were missed or every request was refused.

A prototype requires real personal data: This paper project uses permitted fictional requests and task-relevant inputs.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Students need to return markers correctly and quickly. Test clearer labels before proposing a model.

Independent question

Learners often cannot find borrowed classroom books. Define the problem before choosing an AI tool. Suggest one simple solution to compare with AI.

Expected independent answer

The problem is finding who borrowed a book and where it belongs. Compare a labelled shelf and a checkout record with an AI approach. The solution should address the actual need.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Every classroom problem needs AI.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Time students finding predefined books using each approach, record wrong locations and assistance needed, and use comparable book sets and conditions. Start with the simpler label trial. Choose based on evidence, effort and privacy needs, not the presence of AI.

Explanations for the additional workbook examples

W1 Write the problem without prescribing a tool

Use the ten-request activity log to write a problem statement with a user, setting and observed difficulty. Explain why “Build an AI app” is not yet a problem statement.

For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions. Build an AI app names a solution or construction task without specifying a user difficulty or evidence of need.

W2 Identify claims beyond the evidence

A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

The log establishes four incompatible first choices out of ten supplied fictional requests. It does not establish every pupil’s opinion, the cause of the mismatches or that an untested recommender will improve outcomes. Accept any two of those unsupported conclusions with the bounded observation.

W3 Choose information that serves the outcome

The paper project must suggest an activity compatible with available time and a quiet-room requirement. Choose relevant inputs from minutes available, quiet required, pupil name and home address. State the output and a simpler alternative to a recommender. Explain one boundary.

Minutes available and quiet required are relevant. Name and address are unnecessary. Output a compatible activity suggestion or an explained no-match result. A clearer catalogue or compatibility checklist is a simple alternative. Keep requests fictional and the action advisory; do not extend the task to collecting personal profiles or publishing choices.

W4 Frame a different problem independently

In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

Teams selecting equipment in the supplied exercise had three mismatches out of eight and need a box whose listed contents meet the request, or a clear no-match result. Use requested items and the inventory; personal identities are unnecessary. Exclude real purchases, changing stock or public posting. A clearer inventory or item checklist is a simple alternative. The counts do not explain why teams chose mismatching boxes or prove that a box-finder will improve selection.

W5 Measure help as well as avoided errors

For W4, a proposed box-finder never selects a wrong box because it always reports no match. Explain why counting only wrong suggestions would hide a failure. Name two results to record on test requests and

explain why the original three-of-eight observation is not already an evaluation of the new tool.

Always refusing misses useful matches when suitable boxes exist. Record compatible or incompatible suggestions and whether no-match outputs were justified by the inventory, along with total requests. The three-of-eight observation describes the earlier selection process; the proposed tool has not yet been tested under a comparison procedure.

Assess independent transfer

Selected case: In a fictional equipment cupboard exercise, eight teams choose a box from a list. Three boxes do not contain the items required by the corresponding request; five do. The teacher permits a paper box-finder using fictional item requests and an inventory. Write a brief naming the user difficulty, evidence, desired outcome, necessary information, an excluded action and a simple alternative. Identify one cause or benefit that these counts do not establish.

Selected case answer

Teams selecting equipment in the supplied exercise had three mismatches out of eight and need a box whose listed contents meet the request, or a clear no-match result. Use requested items and the inventory; personal identities are unnecessary. Exclude real purchases, changing stock or public posting. A clearer inventory or item checklist is a simple alternative. The counts do not explain why teams chose mismatching boxes or prove that a box-finder will improve selection.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Choose the Classroom Sorter's real user and measurable need; keep an AI-free solution option open.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 30: Choose Inputs and Outputs. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use sentence starters: In this setting, these users need; the supplied evidence shows; we will check; we will not change. Allow an oral brief recorded by the teacher without supplying the choices. Extension: describe evidence that would make the team abandon its preferred solution.

W4 must transfer the structure: three of eight mismatches, a compatible box or justified no-match outcome, requested items and inventory, one boundary, a simple alternative and a bounded claim. After support use a new scenario: six fictional room requests, two assigned rooms too small, four suitable; permit paper recommendations only. Require capacity and group size as relevant inputs, no real bookings, and no invented explanation for the two failures.

Week 30 turns the activity problem into explicit requirements and a simple baseline for comparison.

Plan the delayed return

Suggested return points in this level: ai-1/30, ai-1/32, ai-1/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Helpful Classroom Sorter, retain the problem record, permitted scope, baseline and observable acceptance criteria.

7. Prepare, teach and assess

Teaching target: Describe a specific user difficulty using supplied evidence without inventing its cause

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A team says that the four mismatches prove every pupil dislikes the catalogue and that a recommender will fix the problem. Identify two unsupported conclusions and state what the log actually establishes.

Reasoned practice answer: The log establishes four incompatible first choices out of ten supplied fictional requests. It does not establish every pupil's opinion, the cause of the mismatches or that an untested recommender will improve outcomes. Accept any two of those unsupported conclusions with the bounded observation.

Check every learner before discussion: Use the worked case for A problem worth solving. Proposed explanation: "Using AI is itself the outcome". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: State what improves for the user and how to observe it.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The approved list contains no tray with every requested tool. Require a prediction and a reason before comparing. Expected result: The finder should report no suitable tray rather than guess. Reason: Success includes honestly identifying unsatisfied requirements.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved

independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.